AQ2.2: Activity Questions 2- Not Graded

This assignment will not be graded and is only for practice.

Level 1:

1 point

Consider a system of linear equations as follows:

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + a_{13}x_3 &= b_1 \\ a_{21}x_1 + a_{22}x_2 + a_{23}x_3 &= b_2 \\ a_{31}x_1 + a_{32}x_2 + a_{33}x_3 &= b_3 \end{aligned}$$

a11x1 + a12x2 + a13x3 = b1a21x1 + a22x2 + a23x3 = b2

a31x1 + a32x2 + a33x3 = b3

Let the matrix representation of the above system be $Ax = b$Ax = b, where $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$A = [a11a21a31a12a22a32a13a23a33], $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$x = [x1x2x3], and $b = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$b = [b1b2b3]. Let A_{x_i} Axi be the matrix obtained by replacing the ii-th column of AA ($i.e.$, $\begin{bmatrix} a_{1i} \\ a_{2i} \\ a_{3i} \end{bmatrix}$)(i.e., [a1ia2ia3i])by bb, for $i = 1, 2, 3$i = 1, 2, 3. If $det(A_{x_i}) = 0$det(Axi) = 0 for $i = 1, 2, 3$i = 1, 2, 3, then which of the following is (are) true?

- [ ] $x = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$x = [000] must be a solution.
- [ ] $x = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$x = [000] is a solution if $det(A) \neq 0$det(A) = 0.
- [ ] If $det(A) = 0$det(A) = 0, then we cannot conclude about the solution using Cramer's rule.
- [ ] If $det(A) \neq 0$det(A) = 0, then $b = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$b = [000].

No, the answer is incorrect.
Score: 0
Accepted Answers:

$x = \begin{bmatrix}0\\0\\0\end{bmatrix}$x = $\begin{bmatrix}0&0&0\end{bmatrix}$ is a solution if $det(A) \neq 0$det(A) = 0.

If $det(A) = 0$det(A) = 0, then we cannot conclude about the solution using Cramer's rule.

If $det(A) \neq 0$det(A) = 0, then $b = \begin{bmatrix}0\\0\\0\end{bmatrix}$b = $\begin{bmatrix}0&0&0\end{bmatrix}$.

Consider a system of linear equations

$$\begin{aligned} x_1 + x_3 &= 1 \\ -x_1 + x_2 - x_3 &= 1 \\ -x_2 + x_3 &= 1 \end{aligned}$$x1 + x3 = 1−x1 + x2 − x3 = 1−x2 + x3 = 1

Let matrix representation of the above system be $Ax = b$Ax = b, where $A = \begin{bmatrix}1&0&1\\-1&1&-1\\0&-1&1\end{bmatrix}$A = $\begin{bmatrix}1&0&1\\-1&0&0\\1&-1&1\end{bmatrix}$, $x = \begin{bmatrix}x_1\\x_2\\x_3\end{bmatrix}$x = $\begin{bmatrix}x1&x2&x3\end{bmatrix}$, and $b = \begin{bmatrix}1\\1\\1\end{bmatrix}$b = $\begin{bmatrix}1&1&1\end{bmatrix}$. Let A_{x_i} Axi be the matrix obtained by replacing the ii-th column of AA (i.e., $\begin{bmatrix}a_{1i}\\a_{2i}\\a_{3i}\end{bmatrix}$ $\begin{bmatrix}a1i&a2i&a3i\end{bmatrix}$) by bb, for $i = 1, 2, 3$i = 1, 2, 3.

(Use the above information to answer questions 2 and 3)

1 point

Choose the set of correct options

☐ $A_{x_1} = \begin{bmatrix}1&1&0\\-1&1&-1\\0&-1&1\end{bmatrix}$Ax1 = $\begin{bmatrix}1&-1&0&1\\1&-1&0&-1\\-1&1&0&-1\end{bmatrix}$

☐ $A_{x_1} = \begin{bmatrix}1&0&1\\1&1&-1\\-1&1&1\end{bmatrix}$Ax1 = $\begin{bmatrix}1&1&1&0\\1&-1&1&-1\\-1&1&-1&-1\end{bmatrix}$

☐ $A_{x_2} = \begin{bmatrix}1&0&1\\-1&1&-1\\0&-1&1\end{bmatrix}$Ax2 = $\begin{bmatrix}1&-1&0&1\\0&0&1&-1\\-1&1&1&-1\end{bmatrix}$

☐

A_{x_3} = [1 0 1 / −1 1 1 / 0 −1 1]Ax3 = [1−1001−1111]

No, the answer is incorrect.
Score: 0
Accepted Answers:

A_{x_1} = [1 0 1 / 1 1 −1 / 1−1 1]Ax1 = [11101−11−11]

A_{x_3} = [1 0 1 / −1 1 1 / 0 −1 1]Ax3 = [1−1001−1111]

1 point
Choose the set of correct options.

☐ x_1 = − 2x1 = −2.

☐ x_2 = − 2x2 = −2.

☐ x_3 = 3x3 = 3.

☐ None of the above.

No, the answer is incorrect.
Score: 0
Accepted Answers:
x_1 = − 2x1 = −2.
x_3 = 3x3 = 3.

Level 2:

Consider the system of linear equations $Ax = b$Ax = b, where A = [1 a 0 / a 1 2 / 1 0 a]A = [1a1a1002a], x = [x_1 / 0 / x_3]

x = [x10x3], and b = [1 / 0]b = [100], and the solution for xx is partially known. What is the value of

a^2 a2, if $a \neq 0, \sqrt{3}, -\sqrt{3}$a = 0, 3

√

, − 3

√

is given? [Hint: Observe that the second row of the vector xx is given as 00, which implies that x_2 x2 is known, where $x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$x = $\begin{bmatrix} \\ \text{x1x2x3} \end{bmatrix}$

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 2
1 point

Consider the system of linear equations $Ax = b$Ax = b, where $A = \begin{bmatrix} 1 & 2 & 3 \\ a+3 & 2a+2 & a+1 \end{bmatrix}$A = [13a + 22a + 31], $x = \begin{bmatrix} x_1 \\ 0 \end{bmatrix}$x = [x10], and $b = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$b = [11], and the solution for xx is partially known.

What is the value of $3a$3a, if $det(A) \neq 0$det(A) = 0 is given?

No, the answer is incorrect.
Score: 0